A Geometric Identity for the Fine-Structure Constant: From the 4D Unit Ball Volume and its Cube-Packing Deficit

(Paper 7: α-Identity)

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Abstract

Let $N(1)$ denote the count of integer-centred unit cubes fully contained in the 4-dimensional ball $B(R)$ of radius $R = 3$. Direct enumeration gives $N(1) = 137$. The volume of the 4D unit ball is $V_4(1) = \pi^2/2$. The combination

$$\alpha^{-1} \stackrel{?}{=} N(1) + V_4(1) \cdot \alpha$$

is a self-consistent quadratic equation

$$\tfrac{\pi^2}{2}\alpha^2 + 137\alpha - 1 = 0$$

whose positive root is $\alpha = 0.0072989\ldots$, agreeing with the observed value $\alpha_{\text{obs}} = 7.2973525 \times 10^{-3}$ to a relative accuracy of 2.1×10^{-4} (0.02%). In this paper we (i) re-derive $N(1) = 137$ and $V_4(1) = \pi^2/2$ by elementary calculation, (ii) verify the numerical validity of the identity, and (iii) propose a structural reading as "tree-level coupling on the inside (packed cubes) plus self-energy correction on the outside (boundary gap)". Whether the identity reflects geometric necessity or numerical coincidence is left as an open problem.

1 Introduction

The fine-structure constant $\alpha = e^2/(4\pi\varepsilon_0\hbar c) \approx 1/137.036$ is among the most precisely measured dimensionless physical constants and governs the strength of the electromagnetic interaction. That its value lies "near 137" has been known since Eddington (1929), and number-theoretic explanations have been attempted for a century. These attempts—from Eddington's naive integer-fitting to Wyler's (1969) multi-element formula—have nearly all been rejected, either by improved measurement precision or by the absence of independent verification.

This paper observes that a problem in 4-dimensional geometry, motivated independently, satisfies a self-consistent equation that reproduces both the integer part 137 and the fractional part 0.036 of α^{-1} using only **parameters that cannot be externally tuned**.

Specifically, from two elementary facts—(i) the classical problem of packing integer-centred unit cubes in a 4D ball (a natural generalisation of the Gauss circle problem to 4D), and (ii)

the volume formula for the 4D ball—an algebraic identity emerges that predicts α^{-1} to 0.02% accuracy.

This paper does not propose a number-theoretic "single integer fit" but expresses α as the **solution of a self-consistent equation**. Since the correction term $V_4(1) \cdot \alpha$ contains α itself, the identity has the structure of a perturbation expansion

$$1 = 137\alpha + V_4(1)\alpha^2,$$

analogous to the renormalisation structure in QFT. This is the decisive difference from naive number-theoretic predecessors.

Structure of the paper. §2 re-derives $N(1) = 137$ and $V_4(1) = \pi^2/2$ independently. §3 performs the numerical verification of the identity. §4 presents a physical reading (inside/outside decomposition). §5 enumerates candidate mechanisms and §6 discusses open problems and the residual. §7 concludes.

The paper is written self-contained: every numerical value is reproducible by elementary calculation in the main text, and references to external papers serve only for context, not for the verification of the main claims.

2 Re-derivation of the Two Geometric Quantities

2.1 $N(1) = 137$: Direct Enumeration

Consider the 4-dimensional ball $B(3) = \{x \in \mathbb{R}^4 : \|x\|_2 \leq 3\}$. An integer-centred cube $C(c) = \prod_i [c_i - 1/2, c_i + 1/2]$ with $c \in \mathbb{Z}^4$ is **fully contained** in $B(3)$ iff every vertex of $C(c)$ lies in $B(3)$, equivalently

$$\sum_{i=1}^{4} (|c_i| + 1/2)^2 \leq 9.$$

Setting $s_i := |c_i| \in \mathbb{Z}_{\geq 0}$, the inequality becomes

$$\sum_i s_i^2 + \sum_i s_i + 1 \leq 9 \quad \Longleftrightarrow \quad \sum_i s_i^2 + \sum_i s_i \leq 8.$$

Enumerating (s_1, s_2, s_3, s_4) in increasing order:

Pattern	$\sum s_i^2 + \sum s_i$	Permutations	Sign DOF	Cube count
$(0,0,0,0)$	0	1	1	1
$(1,0,0,0)$	2	4	2^1	8
$(1,1,0,0)$	4	6	2^2	24
$(1,1,1,0)$	6	4	2^3	32
$(1,1,1,1)$	8	1	2^4	16
$(2,0,0,0)$	6	4	2^1	8
$(2,1,0,0)$	8	12	2^2	48
$(2,1,1,0)$	10	—	—	excluded
$(2,2,0,0)$	12	—	—	excluded
$(3,0,0,0)$	12	—	—	excluded

Permutations count placement choices for non-zero s_i; **sign DOF** is $\pm$ for each non-zero s_i. Total:

$$N(1) = 1 + 8 + 24 + 32 + 16 + 8 + 48 = \boxed{137}.$$

This calculation is fully elementary and rigorous, requiring no external reference.

2.2 $V_4(R) = (\pi^2/2)R^4$ and $V_4(1) = \pi^2/2$

The n-dimensional ball volume formula:

$$V_n(R) = \frac{\pi^{n/2}}{\Gamma(n/2+1)} R^n.$$

For $n = 4$, $\Gamma(3) = 2$ gives

$$V_4(R) = \frac{\pi^2}{2} R^4.$$

In particular $V_4(1) = \pi^2/2 \approx 4.9348022$.

2.3 Unit Value of the 4+1D Schwarzschild–Tangherlini BH Entropy

For reference, the horizon area of the 4+1D Tangherlini (1963) solution is the 3-dimensional measure of a 3-sphere of radius r_h:

$$A_3 = 2\pi^2 r_h^3,$$

and the Bekenstein–Hawking entropy in Planck units ($G_5 = \ell_P^3 = 1$):

$$S_{BH}(r_h) = \frac{A_3}{4} = \frac{\pi^2}{2} r_h^3.$$

At $r_h = 1$:

$$S_{BH}(1) = \frac{\pi^2}{2} = V_4(1).$$

That is, the unit value of the 4+1D BH entropy coincides with the 4D unit ball volume. This follows directly from standard 4D and 4+1D geometric formulae and is not strictly required for the main result, but is used in the physical reading in §4.

3 The Identity and its Numerical Verification

3.1 Main Result

Combining the two quantities $N(1) = 137$ and $V_4(1) = \pi^2/2$ from §2, consider the algebraic identity:

$$\boxed{\alpha^{-1} = N(1) + V_4(1) \cdot \alpha} \tag{1}$$

This is equivalent to a self-consistent quadratic in α:

$$V_4(1) \cdot \alpha^2 + N(1) \cdot \alpha - 1 = 0 \tag{2}$$

$$\frac{\pi^2}{2}\alpha^2 + 137\alpha - 1 = 0.$$

3.2 Solution and Numerical Verification

Positive root of (2):

$$\alpha = \frac{-137 + \sqrt{137^2 + 2\pi^2}}{\pi^2} = \frac{-137 + \sqrt{18788.7392\ldots}}{9.8696044\ldots}.$$

Numerical evaluation:

$$\sqrt{18788.7392} = 137.0720224\ldots$$
$$\alpha = \frac{0.0720224}{9.8696044} = 7.29836 \times 10^{-3}.$$

Reciprocal:

$$\alpha^{-1} = 137.0360\ldots$$

Observed value (CODATA 2018):

$$\alpha_{\text{obs}} = 7.2973525693(11) \times 10^{-3}, \qquad \alpha_{\text{obs}}^{-1} = 137.035999084(21).$$

Quantity	Predicted	Observed	Relative error
α	7.2984×10^{-3}	7.2974×10^{-3}	1.4×10^{-4}
α^{-1}	137.0360	137.0360	$\sim 3 \times 10^{-7}$ (integer part exact)
$\alpha^{-1} - 137$	0.03601	0.03600	3.1×10^{-4}

The integer 137 matches exactly; the fractional 0.036 agrees to within **0.03%**. As α itself, prediction matches observation to **0.02%**.

3.3 Equivalent Forms

(1) admits the equivalent forms

$$1 = 137\alpha + \frac{\pi^2}{2}\alpha^2 \tag{3}$$

which is a power expansion in α: tree-level 137α + first-order correction $V_4(1)\alpha^2$ summing to unity. Physically reminiscent of perturbative expansions in QFT.

Alternative form:

$$137 = \alpha^{-1} - \frac{\pi^2}{2}\alpha = \alpha^{-1}\left(1 - \frac{\pi^2}{2}\alpha^2\right). \tag{4}$$

"Integer 137 = observed α^{-1} minus self-energy term = bare coupling."

4 Physical Reading: Inside/Outside Decomposition

The identity (1) was established as a mathematical fact in §3. This section attempts a physical reading. **Emphasis**: the following interpretation is a structural metaphor, not a first-principles derivation.

4.1 Geometry of the R=3 System

As seen in §2.1, the 4D ball of radius $R = 3$ contains exactly **137 fully inscribed unit cubes**. The ball volume is $V_4(3) = 81 \cdot \pi^2/2 \approx 399.72$. The deficit:

$$\Delta(3) = V_4(3) - N(1) = 81 \cdot \frac{\pi^2}{2} - 137 \approx 262.72.$$

This is the volume of the ball not occupied by full cubes—**65.7%** of the total ball volume. In the R=3 system the boundary dominates over the interior.

Region	Volume	Structure
Interior (137 cubes)	137	discrete, rigid
Boundary (gap)	262.72 (66%)	continuous, surplus
Total (ball)	399.72	$V_4(3)$

4.2 Naturalness of the Inside/Outside Decomposition

Restating (3):

$$1 = 137\alpha + \frac{\pi^2}{2}\alpha^2.$$

We re-read each term:

Term	Physical reading	Origin
137α	inside tree-level coupling	$N(1) = 137$ (packing integer)
$\frac{\pi^2}{2}\alpha^2$	outside self-energy correction	$V_4(1) = \pi^2/2$ (unit ball volume)
Total $= 1$	normalisation of total coupling	—

Physical picture: an electron (or analogous solitonic structure) organises itself at scale $R = 3$. Its 137 internal geometric units carry the tree-level electromagnetic coupling, and its boundary layer (which dominates 66% of the volume) generates a self-correlated renormalisation correction at order α^2.

4.3 Why $V_4(1)$? — Unit Influence-Region Reading

The correction term contains $V_4(R = 1) = \pi^2/2$, not $V_4(R = 3) = 81\pi^2/2$. This suggests that the self-energy correction is a **local process at the unit-cube scale**:

- Each cube correlates with adjacent unit influence regions (4D balls of radius 1, volume $V_4(1) = \pi^2/2$)
- All 137 cubes share the same unit influence region (**intensive**)
- The total correction collapses to a single unit influence-region weight

An equivalent reading: since the unit 4+1D BH entropy $S_{BH}(r_h = 1) = \pi^2/2$ (§2.3) coincides with $V_4(1)$, the correction term can also be read as "**unit-boundary entropy degrees of freedom multiplied by coupling**". These are mathematically the same value with different physical metaphors.

5 Candidate Mechanisms

To physically justify (1) requires an independent derivation of the correction $V_4(1) \cdot \alpha$. We list candidates.

5.1 Candidate A: Vacuum Polarisation (Standard QED Analogue)

QED vacuum polarisation dresses the photon propagator with virtual fermion loops, shifting the low-energy coupling from the bare value:

$$\frac{1}{\alpha(0)} - \frac{1}{\alpha(\Lambda)} = \frac{1}{3\pi} \sum_{\ell} \theta(\Lambda - m_\ell) \left[\ln \frac{\Lambda^2}{m_\ell^2} - \frac{5}{3} \right].$$

In a lattice model, Λ corresponds to a natural cutoff at the lattice spacing. Restricting to the electron loop and demanding the observed shift $\Delta(1/\alpha) = 0.036$, the implied cutoff is $\Lambda \sim 0.6$ MeV—just above m_e. The physical interpretation requires further study, but is consistent with kinematic thresholds at the lightest lepton scale.

5.2 Candidate B: Boundary Self-Correlation

From the inside/outside decomposition in §4.2, where the boundary occupies 66%, a direct mechanism is that the self-correlation energy of boundary modes generates the correction:

$$\delta\alpha^{-1} \sim \int_{\partial} \langle \mathcal{O}(x)\mathcal{O}(y) \rangle \, dx \, dy \stackrel{?}{=} V_4(1) \cdot \alpha.$$

Here $\mathcal{O}$ denotes a boundary field and ∂ the boundary layer of the R=3 ball. Concrete computation is left for future work.

5.3 Candidate C: Beat Coupling with a Longitudinal Scalar Field

Independent thought-experiment work suggests identifying the Higgs (or analogous scalar longitudinal mode) with the longitudinal vibrations of a minimal-phase lattice. **Beat interference** between transverse-wave solitons (e.g. electrons) and the longitudinal Higgs vibration enables small energy transfer between modes that are nominally orthogonal. This mechanism activates when curvature (background geometry) breaks the orthogonality, yielding a correction to the observed α. Details deferred to a separate paper.

5.4 Candidate Comparison

Candidate	Mechanism	Verifiability	Estimated co
A Vacuum polarisation	Virtual lepton pair production	QED-tractable	10^{-3} to
B Boundary self-correlation	Direct inside/outside implementation	Lattice-computable	not yet est
C Scalar longitudinal beat	Thought-experiment based	Conjectural mechanism	not yet est

These are not exclusive; they may represent different facets of the same phenomenon.

6 Open Problems and the Residual

6.1 The 0.03% Residual

As shown in §3.2, identity (1) predicts α^{-1} to 0.03% accuracy. Since experimental precision is $\sim 10^{-10}$, the 0.03% residual is clearly significant. Candidate origins:

1. **Higher-order corrections**: an extension to $1 = 137\alpha + V_4(1)\alpha^2 + c_3\alpha^3 + \cdots$. Reverse-fitting from observed α gives $c_3 \approx 1.4$, but no geometric identification yet.
2. **Corrections to** $V_4(1)$: the basic quantity may itself need higher-order π-corrections (e.g. $\pi^2/2 \to \pi^2/2 + \delta$).
3. **Modification of** $N(1) = 137$: discrete count, so does not admit continuous deformation. Unlikely.
4. **Different mechanism**: identity (1) is approximate; the true equation has a different form.

Quantitatively, the residual is $|0.0360104 - 0.0359991| = 1.13 \times 10^{-5}$. Compared to $V_4(1) \cdot \alpha = 0.036$, the relative residual is $\sim 3 \times 10^{-4}$. This is at the level of one further power of α but larger than $V_4(1) \cdot \alpha^3 \sim 10^{-7}$, suggesting a missing intermediate geometric term.

6.2 The Privilege of $R = 3$

The identity combines a $R = 1$ quantity ($V_4(1)$) and a $R = 3$ quantity ($N(1) = 137$). Why these two scales pair up is unresolved:

- $R = 1$: unit ball (smallest scale)
- $R = 3$: diameter $6 = 2 \times \sqrt{4} =$ (4D unit hypercube diagonal) $\times 3$. Smallest non-trivial complete packing?

In §2.1 the packing problem yields $N(0) = 1$ (trivial), $N(1) = 137$, $N(2) = 1545$, $N(3) = 7281$, ... Why the electron corresponds to $N(1)$ (with possibly the muon at $N(2)$ and tau at $N(3)$ in a generation hierarchy) is for separate study.

6.3 Generation Independence

Electron, muon, and tau share the same charge, so α is generation-universal. The identity predicts a generation-universal α, consistent with this. The mass hierarchy (m_e, m_μ, m_τ) is not addressed and lies outside this paper's scope.

6.4 Relation to the Eddington Trap

As mentioned in §1, Eddington-style number theory failed by attempting to fit α^{-1} with **integers alone**. The present identity:

1. Has the structure **integer 137 + correction containing** α, hence is a self-consistent equation

2. Both the integer 137 and the correction coefficient $\pi^2/2$ are derived **directly and independently from 4D geometry** (§2)
3. The correction has the structural form of a perturbative QFT expansion
4. Candidate physical mechanisms are proposed from a thought experiment (§5)

These are decisive differences from Eddington. Yet whether the identity itself reflects geometric necessity or numerical coincidence remains an open question. This paper does not adjudicate, only presents the observed structure.

7 Conclusion

From two elementary geometric quantities—$N(1) = 137$, the count of integer-centred unit cubes fully contained in a 4D ball of radius 3, and $V_4(1) = \pi^2/2$, the volume of the 4D unit ball—a self-consistent algebraic equation

$$\alpha^{-1} = N(1) + V_4(1) \cdot \alpha$$

is constructed whose solution predicts the observed fine-structure constant to 0.02% accuracy.

This identity is:

1. **Mathematically**: independently verifiable via the elementary calculations in §2 and the numerical check in §3.
2. **Structurally**: of perturbative form $1 = 137\alpha + V_4(1)\alpha^2$, analogous to renormalisation in QFT.
3. **Physically**: readable as inside (137 packed cubes) / outside (66% boundary) decomposition (§4), with multiple candidate mechanisms (§5).

Whether the identity reflects geometric necessity or a mechanism-specific coincidence is unresolved. The contribution of this paper is not a verdict but the presentation of a self-consistent equation, built from 4D geometric quantities not borrowed from elsewhere, that agrees with observation to 0.02%.

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Appendix: Numerical Reproduction (calculator-friendly)

1. $V_4(1) = \pi^2/2 = 9.8696044/2 = 4.9348022$
2. $N(1) = 137$
3. Quadratic $4.9348022\,\alpha^2 + 137\,\alpha - 1 = 0$
4. Discriminant $D = 137^2 + 4 \times 4.9348022 = 18769 + 19.7392 = 18788.7392$
5. $\sqrt{D} = 137.07203$
6. $\alpha = (137.07203 - 137)/(2 \times 4.9348022) = 0.07203/9.8696044 = 7.29836 \times 10^{-3}$
7. $\alpha^{-1} = 137.0360$
8. CODATA 2018: $\alpha^{-1} = 137.035999\ldots$
9. Agreement: relative error $\sim 3 \times 10^{-4}$

The main claim of this paper is fully verifiable on a calculator.